



| 2                 | <p>A headteacher wishes to store details of her students in a computer database. She has been given an example of one student record by the school secretary:</p> <p>Strauss, Johann, SR132516, 83, 01/09/2011</p> <p>This corresponds to family_name, first_name, Student_Id, Maths_test_score, joined_the_school.</p> <p>The file is currently stored in the form of plain text with each record separated by a line break.</p> <p>(a) Give a file extension which is appropriate for this file.</p> <p>.....[1]</p> <p>(b) For each field identify the <b>most</b> appropriate data type. The Maths_test_score is used in calculations.</p> <table><thead><tr><th>Field name</th><th>Data type</th></tr></thead><tbody><tr><td>family_name</td><td></td></tr><tr><td>first_name</td><td></td></tr><tr><td>Student_Id</td><td></td></tr><tr><td>Maths_test_score</td><td></td></tr><tr><td>joined_the_school</td><td></td></tr></tbody></table> <p>[5]</p> <p>(c) Identify the <b>two</b> fields for which a format check would be the <b>most</b> appropriate validation check. In each case describe how the format check would work.</p> <p>1 .....</p> <p>.....</p> <p>.....</p> <p>2 .....</p> <p>.....</p> <p>.....</p> <p>[4]</p> <p>Q14 – Page11 0417/12 May/June 2015</p> | Field name | Data type | family_name |  | first_name |  | Student_Id |  | Maths_test_score |  | joined_the_school |  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|--|------------|--|------------|--|------------------|--|-------------------|--|
| Field name        | Data type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |           |             |  |            |  |            |  |                  |  |                   |  |
| family_name       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |           |             |  |            |  |            |  |                  |  |                   |  |
| first_name        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |           |             |  |            |  |            |  |                  |  |                   |  |
| Student_Id        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |           |             |  |            |  |            |  |                  |  |                   |  |
| Maths_test_score  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |           |             |  |            |  |            |  |                  |  |                   |  |
| joined_the_school |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |           |             |  |            |  |            |  |                  |  |                   |  |

3

A headteacher keeps all details of students in a database.

(a) Describe what is meant by the following terms, using examples of student data.

Field .....

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File .....

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.....

Record .....

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[6]

(b) Describe what is meant by a key field, using examples of student data.

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[3]

4

An airline keeps a database of all the flights it operates. A short extract is shown below.

| Flight_number | Departure_Airport_Code | Number_of_passengers | Destination_Airport_code | Ticket_prices |
|---------------|------------------------|----------------------|--------------------------|---------------|
| EK236         | JNB                    | 135                  | LHR                      | \$1200        |
| EK080         | DXB                    | 256                  | DEL                      | \$500         |
| EK029         | LHR                    | 375                  | DXB                      | \$650         |

(a) Explain fully, what is meant by the following validation checks using the extract above.

Range check .....

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Format check .....

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Length check .....

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[6]

(b) Using the extract above, give an example of **one** field which would contain:

Text data  
(alphanumeric) .....

Numeric data .....

[2]

5

Justdiscs Limited stocks many music CDs. Below is a small selection of CDs stored on their database.

The data has been sorted on two fields.

| Artist_name    | CD_title         | Price   | Year_released | Number_of_tracks |
|----------------|------------------|---------|---------------|------------------|
| Giggs          | Let Em Ave It    | \$10.99 | 2010          | 18               |
| Sabiton        | Heroes           | \$9.99  | 2014          | 12               |
| Luke Slater    | Alright on Top   | \$8.99  | 2001          | 10               |
| Arctic Monkeys | Humbug           | \$8.99  | 2009          | 12               |
| Oasis          | Definitely Maybe | \$8.99  | 2014          | 11               |
| Erasure        | Erasure          | \$7.99  | 2009          | 11               |
| Jimmy Smith    | Cool Blues       | \$4.99  | 2002          | 7                |

- (a) Write down the field which was used as the primary sort in the database and the order in which it was sorted.

.....  
 ..... [2]

- (b) Write down the field which was used as the secondary sort in the database and the order in which it was sorted.

.....  
 ..... [2]

- (c) Peter is the owner of the shop. He receives requests from customers about the CDs he has in stock. He converts these requests into search criteria using comparison and Boolean operators.

For example, one customer might ask for CDs which were released after 2002 and also those that have at least 11 tracks. Peter would write this down as:

Year\_released > 2002 **OR** Number\_of\_tracks >= 11

Write down the search criteria for a customer who wants a list of all the CDs which were released before 2010 but only if they cost less than \$8.99

..... [5]

- (d) Write down the names of the artists whose records match the requirements of part (c).

..... [2]

**Q10 – Page 7 0417/11 May/June 2016**

6

Maria is the principal of the International Municipal school. She has employed Paulo, a systems analyst, to create a new database system to store records of her final year IGCSE students.

Examples of the details of the students which will be stored are:

Velia Grimaldi, A3058, Female, 161, Leaving

Giuseppe Campo, A3072, Male, 177, Staying on to 6th form

Nicola Donati, B3085, Female, 173, Staying on to 6th form

Giovanni Agnelli, C3102, Male, 172, Leaving

- (a) Complete the following table by entering the field names and **most appropriate** data type for each field.

For any numeric field, specify the type of number.

| Field name | Data type |
|------------|-----------|
| Name       |           |
|            |           |
|            | Boolean   |
| Height     |           |
|            |           |

[7]

- (b) Paulo wishes to save storage space. He does not change the Name field, but decides that two other fields could have their data shortened.

Apart from the Name field, identify the two other fields and describe how the contents will be shortened.

Field 1.....

Shortened form 1.....

Field 2.....

Shortened form 2.....

[4]

(c) There is no key field set. Maria's secretary typed in the data to the database in this order:

Giuseppe Campo, A3072, Male, 177, Staying on to 6th form

Giovanni Agnelli, C3102, Male, 172, Leaving

Nicola Donati, B3085, Female, 173, Staying on to 6th form

Velia Grimaldi, A3058, Female, 161, Leaving

Maria can get the database back in to the original order by sorting the data using one field.

Write down the name of this field and the order in which it will need to be sorted.

Field.....

Order.....

**Q12 – Page 8 0417/12 May/June 2016 [2]**

7

Enrique has developed a new database system for a supermarket. Below is one record from the database. He needs to test the validation checks that he has developed for the system.

| Barcode       | Supplier_code | Contents       | Price | Reorder_level | Weight(kg) |
|---------------|---------------|----------------|-------|---------------|------------|
| 5012427141308 | EZ123         | Lobster Bisque | £2.72 | 100           | 0.4        |

(d) For each field write down the **most appropriate** data type. For any numeric field specify the type of number.

Barcode.....

.....

Supplier\_code.....

.....

Contents.....

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Price.....

.....

Reorder\_level.....

.....

Weight(kg).....

.....

[6]

**Q11 – Page 8 - d 0417/13 May/June 2016**

8

A school's examinations officer has decided to develop a database to store pupil details and the examinations they are due to sit next summer. They are going to use a relational database.

(a) Describe the steps involved in creating the database.

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[4]

(b) Some of the records stored in the database are given below.

| Examination_ID | Name_of_subject       | Examination_level | Date_of_exam |
|----------------|-----------------------|-------------------|--------------|
| 041731         | ICT                   | IGCSE             | 12/04/2018   |
| 047822         | Computer Science      | IGCSE             | 11/06/2018   |
| 604312         | Design and Technology | O Level           | 01/06/2018   |
| 301532         | French                | O Level           | 28/06/2018   |

Based only on the examination details above, complete the table below, stating the most appropriate **data type** for each field.

| Field name        | Data type |
|-------------------|-----------|
| Examination_ID    |           |
| Name_of_subject   |           |
| Examination_level |           |
| Date_of_exam      |           |

[4]



- (c) Write the search criteria to search for all examinations which will be sat after 26/06/2018, using at least one of the following operators >, >=, <, <=, = .

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..... [2]

- (d) Apart from presence check, name and describe a suitable validation check for each of the following fields in the database, based on the data shown in **part (b)**.

Examination\_ID.....

.....

.....

Date\_of\_exam.....

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.....

[4]

**Q9 – Page 7 0417/12 May/June 2017**

**9**

A club is setting up a database to store details of its members.

- (a) Complete the data dictionary of the database, adding the most appropriate data type for each field.

| Field name                | Data type |
|---------------------------|-----------|
| Membership_number         | Number    |
| Member_name               |           |
| Membership_fee_paid       | Boolean   |
| Telephone_number          |           |
| Expiry_date_of_membership |           |
| Membership_fee_amount     |           |

[4]

- (b) The club secretary wishes to find all the junior members who have paid their membership fee which would be less than \$100.

He searches for these members by typing in the following search criteria.

**Membership\_fee\_amount < 100 AND Membership\_fee\_paid ="Y"**

He now wishes to invite selected members to a special evening. These members will be those with a membership number over 200 as well as members whose membership expiry date is before 1 Jan 2017.

Write search criteria to list all the members that should be invited.

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 ..... [5]

- (c) Discuss the benefits and drawbacks of storing membership details on a computer database.

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 .....  
 .....  
 .....  
 ..... [6]

**Q8 – Page 8 0417/13 May/June 2017**

10

A farmer has purchased a computerised milking system for her cows. She has asked a systems analyst to create a database to store details of the cows being milked.

Examples of the details of the cows which will be stored are:

| Breed_of_cow | Date_of_birth | Weight_of_cow | Average_milk_yield | Animal_passport_number |
|--------------|---------------|---------------|--------------------|------------------------|
| Holstein     | 25/02/2017    | 725.9         | 24.5               | 998/2017               |
| Ayrshire     | 15/03/2016    | 715.0         | 20.1               | 972/2016               |
| Jersey       | 25/02/2017    | 732.7         | 25.0               | 971/2016               |
| Holstein     | 10/10/2016    | 715.0         | 25.0               | 765/2016               |

(a) Complete the following table by entering the most appropriate data type for each field.

For any numeric field, specify the type of number.

| Field name             | Data type |
|------------------------|-----------|
| Breed_of_cow           |           |
| Date_of_birth          |           |
| Weight_of_cow          |           |
| Average_milk_yield     |           |
| Animal_passport_number |           |

[5]

(b) State which field would be the most appropriate for the primary key.

..... [1]

(c) A validation check is used when entering the animal passport number into the database.

Name and describe the most appropriate validation check that could be applied to this field.

.....  
 .....  
 .....  
 ..... [2]

**Q9 – Page 7 0417/11 May/June 2018**



12

Tawara Health Centre is creating a new online appointment system for its patients.

The developer has created a table showing part of the file structure of the new database.

| Field name               | Data type        | Field length | Comment                           |
|--------------------------|------------------|--------------|-----------------------------------|
| Medical_ID_number        | Numeric: integer |              | 12-digit unique number            |
| Date_of_birth            | Date/Time        |              | dd/mm/yy                          |
| Family_name              | Text             | 10           |                                   |
| Doctor_name              | Text             | 20           |                                   |
| Appointment_time         | Date/Time        | 5            | hh:mm                             |
| Contact_telephone_number | Text             |              | 01617151234                       |
| Current_medication       | Text             | 20           |                                   |
| Blood_type               | Text             | 3            | A, O, AB, B<br>Each can be + or – |

- (a) Apart from presence check, for each field, name an appropriate validation check that could be used. Your answer should be different in each case.

| Field name               | Validation check |
|--------------------------|------------------|
| Medical_ID_number        |                  |
| Blood_type               |                  |
| Date_of_birth            |                  |
| Contact_telephone_number |                  |

[4]

- (b) Referring to the fields in the table, name the most appropriate field that could be used as a primary key.

..... [1]

**Q8 – Page 6 a-b 0417/13 May/June 2019**

13

A hospital is creating a database to store blood donors' medical details. The fields in the database include Health\_number, Date\_of\_birth, Gender and Blood\_group.

(a) When a new blood donor wishes to give blood they need to fill in a form.

Each blood donor has a unique health number which uses a 10-digit number, for example 9434765919.

Blood group can be A, B, AB or O.

Part of the form is shown below.

|                                |                                                                                                                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Blood donor                    |                                                                                                                                                                                                                   |
| 1. What is your health number? | <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/> |
| 2. What is your date of birth? | <input type="text"/> <input type="text"/> / <input type="text"/> <input type="text"/> / <input type="text"/> <input type="text"/> <input type="text"/> <input type="text"/>                                       |
| 3. What is your gender?        | M/F                                                                                                                                                                                                               |
| 4. What is your blood group?   | <input type="text"/>                                                                                                                                                                                              |

Name a suitable data type for the answers given to each of the questions in the form. For any numeric field, specify the type of number.

Question 1.....

Question 2.....

Question 3.....

Question 4.....

[4]

(b) In the Blood\_group field the only data that can be entered is A, B, AB or O.

Tick (✓) whether each of the following blood groups is an example of **abnormal** or **normal** data.

| Blood group | abnormal<br>(✓) | normal<br>(✓) |
|-------------|-----------------|---------------|
| AB          |                 |               |
| C           |                 |               |
| B           |                 |               |
| AO          |                 |               |

[2]

(c) The administrator needs a list of all the blood donors in the database with blood group A as well as in the same list all the blood donors with blood group AB. This list will only include female blood donors.

To find the blood donor with the health number 9434765919 he will need to type in the following query.

Health\_number = 9434765919

Write a query to find all the female blood group donors with the blood group A as well as those with AB.

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[6]

(d) When the details of a new donor are added to the database the person’s date of birth is entered.

Explain why the administrator would prefer their date of birth to be stored rather than their age.

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[4]

(e) Name the most appropriate validation check for the Health\_number.

.....

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[1]



14

The 2020 Olympic Games committee is producing a database to show the medal winners in the cycling events.

It is important that the correct data type is used in each of the fields.

(a) Tick (✓) the most appropriate data type for each of the following fields.

|                  | Boolean<br>(✓) | Numeric<br>(✓) | Text<br>(alphanumeric)<br>(✓) |
|------------------|----------------|----------------|-------------------------------|
| Athlete_name     |                |                |                               |
| Gold_medal (Y/N) |                |                |                               |
| Age              |                |                |                               |
| Name_of_country  |                |                |                               |

[4]

Q1 – Page 2 0417/12 May/June 2020

15

The secretary of the Tawara Rowing Club is organising a presentation evening for its members. A systems analyst is creating a database for the club to store the details of the members. She is also creating a spreadsheet to show who is attending the presentation evening.

The systems analyst has set up the following fields for the database.

| Name_of_person | Date_of_birth | Membership_type | Contact_email     | Contact_phone_no | Year_joined | Gender |
|----------------|---------------|-----------------|-------------------|------------------|-------------|--------|
| Nor Kwa        | 2/4/2005      | Social          | n.kwa@rockict.com | 03 2453 5673     | 2018        | F      |
| Adam Mazian    | 23/5/2003     | Social          | AdamM@abc.co.my   | 082 25 4689      | 2016        | M      |
| Ahmed Othman   | 12/03/2006    | Junior          | AOthman@cie.org   | 01223 432678     | 2018        | M      |
| Zara Png       | 1/12/1997     | Senior          | Zara@cbc.cn       | 123 3267 9999    | 2010        | F      |

- (a) For the following fields write down the most appropriate data type. Each data type must be different. For any numeric field, specify the type of number.

Gender.....

Membership\_type.....

Year\_joined.....

[3]

Q6 – Page 6 0417/13 May/June 2020